

HR ANALYTICS DATASET

Power BI End-to-End Project — Student Guide

This guide accompanies the five-table HR Analytics practice dataset (Employees, Departments, Roles, Locations, EmployeePerformance). It explains the data model, documents the data quality issues built into the files on purpose, and gives a structured path for cleaning, modeling, and reporting the data in Power BI.

Dataset files	5 CSV files (linked by ID columns)
Employee records	1,000 unique employees (1,015 rows incl. duplicates)
Performance reviews	3,013 unique reviews (3,043 rows incl. duplicates)
Departments / Roles / Locations	8 / 15 / 5
Time span	Hire dates from June 2021 to June 2026 (5 years)
Intended audience	Beginner to intermediate Power BI learners
Core skills practiced	Power Query cleaning, star-schema modeling, DAX, dashboard design

Note — this dataset is entirely synthetic. Names, salaries, and contact details are randomly generated and do not represent real people or organizations.

1 Project Overview

The goal of this project is to take a realistic but imperfect HR dataset and turn it into a clean, well-modeled Power BI report. The raw files mimic exports you would typically get from an HRMS system — multiple related tables, inconsistent text entry, missing fields, and a few outliers. None of this is random noise to ignore; cleaning and validating it is the main point of the exercise.

By the end of this project you should be able to:

- Profile and clean multi-table data in Power Query (missing values, casing, spacing, mixed date formats, duplicates).
- Build a star-schema data model with correct one-to-many relationships.
- Write DAX measures for HR metrics such as attrition rate, average tenure, and promotion rate.
- Design a multi-page Power BI report that answers specific HR business questions.

Files included

- **Employees.csv** — one row per employee record (fact-like dimension; also the hub of the model).
- **Departments.csv** — department lookup table.
- **Roles.csv** — job role / level lookup table.
- **Locations.csv** — office city lookup table.
- **EmployeePerformance.csv** — one row per annual performance review per employee (fact table).

2 Data Model

The five tables form a star schema. Employees sits at the centre, with three dimension tables (Departments, Roles, Locations) feeding into it on a one-to-many basis, and one fact table (EmployeePerformance) hanging off it on a many side. Employees also contains a self-referencing ManagerID column for org-hierarchy analysis — this is optional to model and is not required for the core relationships.

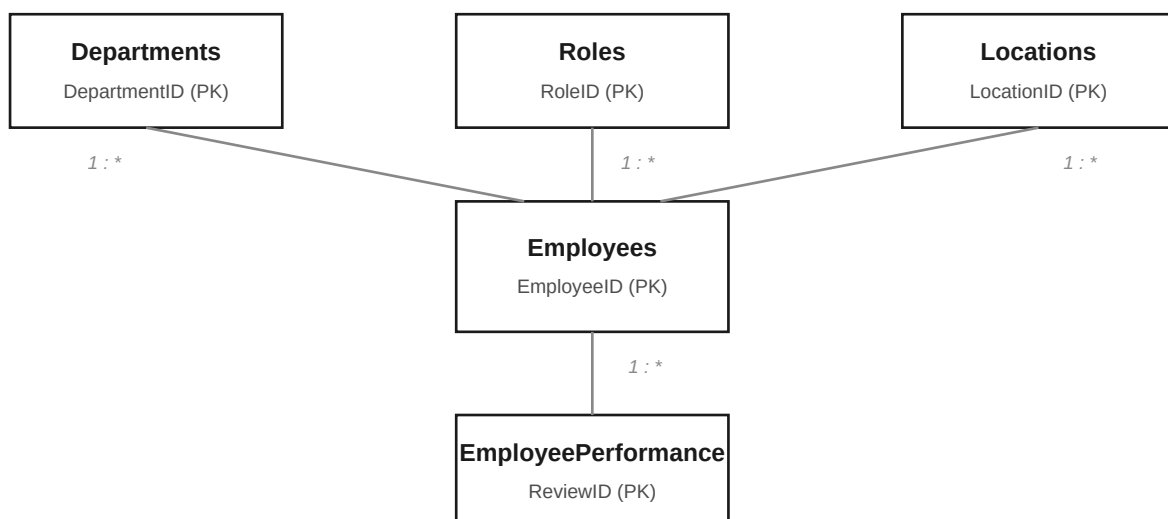


Figure 1 — star schema. Arrows point from the “1” (lookup) side to the “many” (fact) side of each relationship.

From (1 side)	To (many side)	Join column	Notes
Departments	Employees	DepartmentID	Every employee belongs to exactly one department.
Roles	Employees	RoleID	Every employee holds exactly one job role.
Locations	Employees	LocationID	Every employee is based at exactly one office city.
Employees	EmployeePerformance	EmployeeID	Each employee has one review row per year since hire.
Employees	Employees (self)	ManagerID → EmployeeID	Optional. Use for org-hierarchy visuals only; do not use as a core star-schema relationship.

Tip — also build a separate Date table (Calendar) from HireDate / ReviewDate and mark it as a date table. This is required for any time-intelligence DAX (YTD, same period last year, rolling 12 months).

3 Table & Column Reference

3.1 Employees.csv

Grain: one row per employee. Approx. 1,000 unique employees, 1,015 raw rows (includes ~1.5% duplicate rows for cleaning practice).

Column	Type	Description
EmployeeID	Text (PK)	Unique employee code, e.g. E0108.
EmployeeName	Text	Full name. Some rows have extra spaces or inconsistent casing.
Gender	Text	Male / Female, with inconsistent values to standardize (see Section 4).
Age	Whole number	Employee age in years. Some values missing.
DepartmentID	Whole number (FK)	Links to Departments.DepartmentID.
RoleID	Whole number (FK)	Links to Roles.RoleID.
LocationID	Whole number (FK)	Links to Locations.LocationID.
HireDate	Text (mixed date formats)	Date employee joined. Stored in four different text formats — must be parsed.
Salary	Whole number	Annual salary in INR. Contains missing values and outliers.
MaritalStatus	Text	Single / Married / Divorced. Some blank or lower-cased.
EducationLevel	Text	Highest qualification. Casing/spacing inconsistent in places.
PerformanceRating	Whole number (1–5)	Latest overall rating. Some values missing.
AttritionStatus	Text	Yes / No, with inconsistent casing to standardize.
ManagerID	Text (self FK, nullable)	EmployeeID of the employee's manager. Blank for senior leadership.

Column	Type	Description
Email	Text, nullable	Work email. Some blank, some malformed.
PhoneNumber	Text, nullable	Contact number. Some blank, some malformed (wrong length).

3.2 Departments.csv

Grain: one row per department. 8 departments (9 raw rows, including 1 duplicate row).

Column	Type	Description
DepartmentID	Whole number (PK)	Unique department code, 1–8.
DepartmentName	Text	Department name. Spelling/casing/spacing intentionally inconsistent.
DepartmentHead	Text	Name of the department head.
Budget	Whole number	Annual department budget in INR.

3.3 Roles.csv

Grain: one row per job role. 15 roles.

Column	Type	Description
RoleID	Whole number (PK)	Unique role code, 1–15.
JobRole	Text	Job title, e.g. Data Analyst, Sales Executive. A few rows have spacing/casing issues.
JobLevel	Text	Junior / Mid / Senior / Manager.
ExpectedSalaryRange	Text	Reference salary band for the role, formatted as “min-max” in INR.

3.4 Locations.csv

Grain: one row per office city. 5 locations.

Column	Type	Description
LocationID	Whole number (PK)	Unique location code, 1–5.
City	Text	Office city (Hyderabad, Bangalore, Mumbai, Delhi, Pune). Casing/spacing inconsistent in places.
State	Text	State the city is in.
Country	Text	Country name; casing inconsistent in places (“India” / “india”).

3.5 EmployeePerformance.csv

Grain: one row per employee per review year. Approx. 3,013 unique reviews, 3,043 raw rows (includes ~1% duplicate rows).

Column	Type	Description
ReviewID	Text (PK)	Unique review code, e.g. PR02029.
EmployeeID	Text (FK)	Links to Employees.EmployeeID.
ReviewDate	Text (mixed date formats)	Date of the review. Same four mixed formats as HireDate.
PerformanceScore	Decimal (1–5)	Review score for that year. Some values missing.
TrainingHours	Whole number	Training hours completed that year. Some values missing.
OvertimeHours	Whole number	Overtime hours logged that year. Contains a few outliers.
PromotionEligible	Text	Yes / No, with inconsistent casing to standardize.
SatisfactionScore	Whole number (1–5)	Employee satisfaction score for that year. Some values missing.

4 Known Data Quality Issues

These issues were added on purpose so you can practice real cleaning techniques in Power Query. Treat this table as a checklist, not an exact answer key — find and verify each issue yourself before fixing it.

Issue type	Where it appears	What to do
Mixed date formats	Employees.HireDate, EmployeePerformance.ReviewDate (DD-MM-YYYY, MM/DD/YYYY, YYYY-MM-DD, DD-Mon-YYYY all mixed in the same column)	Parse with a custom column / conditional logic, or split-clean by format pattern, then convert to a single Date type.
Missing values	Age, Salary, HireDate, MaritalStatus, PerformanceRating, ManagerID, PerformanceScore, TrainingHours, SatisfactionScore (roughly 2–7% of rows per column)	Decide a sensible fill rule per column (e.g. median for numeric, “Unknown” for category) or leave blank and handle in DAX with care.
Inconsistent text casing / spacing	Gender (Male/M/male/MALE), AttritionStatus (Yes/No/yes/YES), EducationLevel, MaritalStatus, DepartmentName, City, Country, EmployeeName, JobRole	Trim() and Clean() whitespace, then standardize casing and map abbreviations to full values.
Inconsistent department naming	Departments.DepartmentName (e.g. “I.T.” vs “Information Technology”, “HR ” with trailing space)	Clean this dimension table first — everything downstream depends on it.
Null contact details	Employees.Email, Employees.PhoneNumber (roughly 7% blank each)	Leave as null where genuinely missing; do not fabricate values. Flag with an “Is Missing” column if needed for analysis.
Malformed contact details	A few Email values missing the @ symbol; a few PhoneNumber values with the wrong digit count	Validate format with conditional logic; correct or null out invalid entries.
Salary outliers	Employees.Salary — a small number of values far above (Rs. 50L–Rs. 72L) or below (Rs. 8K–Rs. 15K) the realistic range	Identify with a box plot / IQR check before deciding whether to cap, exclude, or investigate.
Duplicate records	Employees.csv (~15 duplicate rows), EmployeePerformance.csv (~30 duplicate rows), Departments.csv (1 duplicate row)	Remove exact duplicate rows; keep the first occurrence by ID.

5 Suggested Power Query Cleaning Checklist

Work top-down through the dimension tables before touching the fact tables.

- **Departments** — Trim/Clean DepartmentName, standardize casing, remove the duplicate row, confirm 8 unique departments remain.
- **Roles** — Trim/Clean JobRole, confirm consistent casing, split ExpectedSalaryRange into Min/Max columns if needed for reference.
- **Locations** — Trim/Clean City and Country, standardize casing.
- **Employees** — Remove duplicate rows; standardize Gender and AttritionStatus to two clean values; Trim/Clean text columns; parse HireDate into a real Date type; decide a handling rule for missing Age, Salary, MaritalStatus, PerformanceRating; review Salary outliers; validate Email and PhoneNumber formats.
- **EmployeePerformance** — Remove duplicate rows; parse ReviewDate into a real Date type; decide a handling rule for missing scores and hours; review OvertimeHours outliers.
- **Relationships** — Load all five tables, build the one-to-many relationships shown in Section 2, and create a separate Date table for time intelligence.

Validation checks before moving to DAX

- Row counts: Departments = 8, Roles = 15, Locations = 5, Employees = 1,000 unique, EmployeePerformance ≈ 3,013 unique.
- Every DepartmentID, RoleID and LocationID in Employees exists in its lookup table (no orphan keys).
- Every EmployeeID in EmployeePerformance exists in Employees.
- Gender has exactly 2 distinct values; AttritionStatus has exactly 2 distinct values.
- All date columns are of Date type, not text.

6 Suggested DAX Measures

A starting set of measures, grouped by business scenario. Adjust table and column names if you renamed anything during cleaning.

Headcount & Attrition

```
Total Employees = DISTINCTCOUNT ( Employees[EmployeeID] )

Active Employees = CALCULATE ( [Total Employees], Employees[AttritionStatus] = "No" )

Attrition Count = CALCULATE ( [Total Employees], Employees[AttritionStatus] = "Yes" )

Attrition Rate % = DIVIDE ( [Attrition Count], [Total Employees] )
```

Salary & Compensation

```
Average Salary = AVERAGE ( Employees[Salary] )

Median Salary = MEDIAN ( Employees[Salary] )

Total Salary Cost = SUM ( Employees[Salary] )
```

Tenure

```
Average Tenure (Years) =
AVERAGEX ( Employees, DATEDIFF ( Employees[HireDate], TODAY (), YEAR ) )
```

Performance & Promotion

```
Average Performance Score = AVERAGE ( EmployeePerformance[PerformanceScore] )

Promotion Eligible Count =
CALCULATE ( COUNTROWS ( EmployeePerformance ), EmployeePerformance[PromotionEligible] =
"Yes" )

Promotion Rate % = DIVIDE ( [Promotion Eligible Count], COUNTROWS ( EmployeePerformance
) )
```

Training & Engagement

```
Average Training Hours = AVERAGE ( EmployeePerformance[TrainingHours] )

Average Overtime Hours = AVERAGE ( EmployeePerformance[OvertimeHours] )

Average Satisfaction Score = AVERAGE ( EmployeePerformance[SatisfactionScore] )
```

Gender Diversity

```
Female Employees = CALCULATE ( [Total Employees], Employees[Gender] = "Female" )

Female Ratio % = DIVIDE ( [Female Employees], [Total Employees] )
```

7 Suggested Report Pages

Map each business scenario to its own report page (or a tab within one page). Suggested visuals are a starting point — adapt them once you see the cleaned data.

Business scenario	Suggested visuals
Employee Attrition Analysis	KPI cards (headcount, attrition rate); attrition by department, role, and tenure band; trend over hire year.
Department Performance Analysis	Average performance score and headcount by department; budget vs. headcount.
Salary Analysis	Average/median salary by role and department; salary distribution (box plot or histogram); pay range vs. ExpectedSalaryRange.
Employee Satisfaction Analysis	Average satisfaction score by department and tenure; satisfaction vs. overtime hours scatter.
Promotion Analysis	Promotion eligibility rate by department and performance band; eligible-but-not-promoted gap.
Training Effectiveness Analysis	Average training hours vs. performance score; training hours by department.
Gender Diversity Analysis	Headcount and average salary by gender; gender mix by department and role; diversity trend over hire year.
Workforce Distribution Analysis	Headcount by location, department, and job level; map or matrix view by city.

8 Step-by-Step Project Workflow

- 1 Import all five CSV files into Power BI (Get Data → Text/CSV).
- 2 Profile each table in Power Query (Column Quality / Column Distribution / Column Profile) to confirm the issues in Section 4.
- 3 Clean each table following the checklist in Section 5, starting with the dimension tables.
- 4 Load the cleaned tables into the data model and build the relationships shown in Section 2.
- 5 Create a Date table and mark it as a date table; relate it to HireDate and/or ReviewDate as needed.
- 6 Build the DAX measures in Section 6 inside a dedicated measures table.
- 7 Build one report page per business scenario from Section 7, starting with KPI cards before adding detail visuals.
- 8 Add slicers for Department, Role, Location, Gender, and Hire Year so every page is filterable.
- 9 Sanity-check the numbers: do headcounts, attrition rate, and average salary look realistic after cleaning?
- 10 Write a short README or notes page summarizing the cleaning decisions you made and why.

End of guide. This dataset and guide are intended for practice purposes only.